



A survey on semiconductor wafer yield prediction by artificial intelligence

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ABSTRACT

Wafer manufacturing is a highly complex process, typically involving hundreds of steps. Yield, which represents the number of fully functional chips on a wafer, is directly related to the final profitability of wafer manufacturing. Utilizing artificial intelligence for yield prediction offers numerous benefits and has garnered extensive research attention. This paper categorizes and examines various algorithms used in yield prediction. While machine learning algorithms exhibit unique advantages when applied to tabular data, numerous deep learning algorithms have also been applied in recent years. We analyze different datasets and algorithms, suggesting that further research in yield prediction can focus on few-shot analysis, model transparency, and multi-modal prediction to achieve earlier and more accurate yield predictions. Through the study and discussion of recent advancements in wafer yield prediction, this paper aims to provide a reference for future research directions in yield prediction.

1. Introduction

Semiconductor manufacturing is a complex and highly sophisticated process involving multiple steps, including wafer fabrication, lithography, doping, wafer acceptance test (WAT), circuit probing (CP) test, packaging and final test (FT) [1], as shown in Fig. 1. Each step can impact the final product's quality and performance [2]. A successful semiconductor manufacturing technology must achieve sufficiently high yields [1]. Yield, a frequently used evaluation metric in semiconductor manufacturing, is the ratio of the number of chips that pass testing after completing all process steps to the number of active chips on the entire wafer [3]. Higher yields indicate higher manufacturing proficiency and potentially greater profitability [2].

The impact of output prediction in semiconductor manufacturing industry is very significant, which is directly related to the profitability and production efficiency of enterprises [4]. Accurate output prediction can help enterprises optimize production processes, reduce waste, improve yield, and improve overall economic benefits. In wafer manufacturing, yield prediction can help identify and resolve potential problems in the production process, thereby reducing scrap rates. For example, one wafer fabrication plant significantly improved the economics of its production line by improving its yield prediction model and reducing the scrap rate from 5% to 2%. In recent years, the wafer yield prediction technology continues to develop, and deep learning

(DL) is widely used in yield prediction, such as convolutional neural network (CNN) and recurrent neural network (RNN), which have made remarkable progress in yield prediction. They are able to process large amounts of high-dimensional data and capture complex patterns and relationships, thereby improving the accuracy of predictions. With the development of Internet of Things (IoT) technology, real-time data acquisition and processing are becoming possible. Manufacturing enterprises can monitor the production process in real time, adjust the production parameters in time, and improve the yield. For example, with real-time predictive models, some companies are able to adjust process parameters in real time during production, avoiding potential yield problems [5].

However, integrated circuits (ICs) on a wafer are susceptible to defects arising from various process steps during the complete manufacturing process [6]. When a defect occurs in a defect-sensitive region, it is referred to as a fatal defect or fault, directly leading to reduced yield [6]. Since yield calculation is performed after all process steps are completed, it serves as a proxy for assessing the feasibility and efficiency of the manufacturing process [7]. Therefore, the ability to predict yield quickly is essential. Predicting yield in advance makes it possible to adjust process parameters to improve yield and reduce losses [8].

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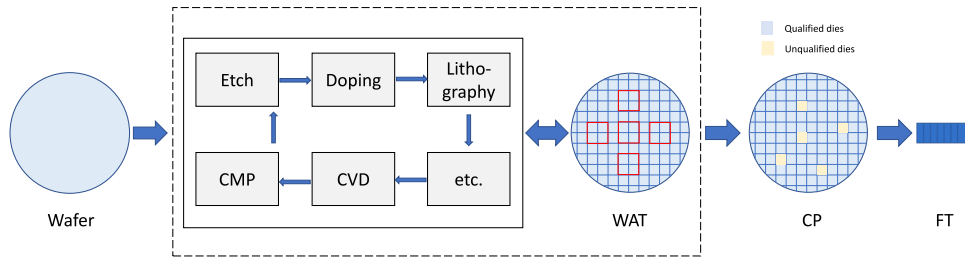


Fig. 1. Wafer manufacturing flow.

With technological advancements, yield prediction methods have evolved. Early yield prediction was primarily based on statistical methods [7]. For example, as early as 1981, Hemmert et al. [9] divided wafers into several Poisson regions with varying defect densities to simulate the yield-area relationship and predict yield. However, due to the limitations of statistical methods in simulating non-linear data in multi-dimensional cases, researchers have shifted from statistical methods to yield prediction methods within the framework of computer-supported ML [10]. Currently, mainstream wafer yield prediction methods include classification, regression, and clustering prediction methods based on ML frameworks [10].

Over the past few decades, various yield prediction methods have emerged. Integrating various mathematical theories and blurring traditional ML and DL frameworks have made categorizing these methods extremely complex [11]. For example, Kumar et al. [12] reviewed traditional yield models, but yield prediction methods have evolved into ML methods in the context of big data. A comprehensive review of all articles using AI methods for yield prediction is necessary to study wafer yield prediction better.

This paper systematically reviews recent research on wafer yield prediction using ML and DL methods, categorizing the prediction approaches into two major classes: ML-based methods and DL-based methods. We conduct a detailed analysis of the datasets currently used by these models and thoroughly examine the advantages and disadvantages of various ML and DL techniques. Additionally, we explore future directions for yield prediction development. The paper identifies several key challenges in current yield prediction research: (1) Dataset Collection Issues: As process nodes shrink and wafer costs rise, collecting large-scale datasets for products under development becomes increasingly difficult, posing significant limitations for big data-driven ML methods. (2) Black-Box Model Limitations: Most existing yield prediction models rely on complex black-box approaches, lacking interpretability in decision-making—a critical drawback for semiconductor manufacturing, where physical mechanisms are highly intricate and production costs are prohibitively high. (3) Single-Data Focus: The majority of yield prediction studies are based on single data types, making it challenging to fully characterize the complex manufacturing process. By reviewing the evolution and current state of yield prediction research, this paper proposes future development directions, including small-sample prediction, interpretable models, and multi-modal fusion approaches. Not only does this study provide a comprehensive overview of wafer yield prediction research, but it also offers constructive insights for future advancements in the field.

The structure of this paper is as follows: Section 2 introduces the background of yield prediction, including the definition of yield and an overview of datasets used in wafer manufacturing. Section 3 provides an overview of traditional ML methods. Section 4 outlines DL schemes. Section 5 discusses prospects for the development of yield prediction methods. Finally, Section 6 summarizes the key points discussed in this paper.

2. Background

2.1. Yield introduction

Yield is a critical indicator in manufacturing and production that measures the proportion of qualified products in the production process [6]. It is commonly used to assess the proportion of qualified products in a production batch to understand the efficiency of the production process and the effectiveness of quality control [12]. High yield means less waste is produced in the production process, the process is stable, and the quality is high [9]. The following formula defines the yield [12]:

$$\text{Yield} = \frac{\text{Qualified.quantity}}{\text{Total.production.quantity}} \times 100\% \quad (1)$$

The quantity of qualified products refers to the number of products that meet quality standards and requirements. The total production quantity refers to the total number of products, including qualified and unqualified products. Yield reflects the efficiency of the production process, the level of process control, and the effectiveness of quality management [3]. A higher yield indicates a more stable production process and effective quality control, reducing production costs and resource waste [12].

2.2. Data introduction

All yield prediction methods require data to support them. However, no existing article clearly explains the forms and functions of various data types. This paper introduces the most commonly used data forms in semiconductor manufacturing, including Fault Detection and Classification (FDC), WAT, Process Parameter, Metrology, CP, and FT data.

2.2.1. FDC data

FDC data is collected from the FDC system, a key component of wafer fabrication process control [13]. It monitors and analyzes equipment operating status and process parameters to promptly address potential failures. FDC systems minimize equipment downtime and production interruptions through real-time monitoring and fault warnings, enhancing wafer productivity and yields [14]. FDC data provides feedback on system characteristics such as temperature, pressure, humidity, and gas flow rate. Statistical representations recorded include minimum, maximum, mean, standard deviation, and process variation [15].

2.2.2. WAT data

WAT data is derived from tests conducted on wafers to evaluate electrical parameters, physical dimensions, and defects [16]. It monitors semiconductor process parameters and typically includes hundreds or thousands of tests (e.g., threshold voltage, leakage current, line width, defect density). WAT data reveals electrical defects and failure modes, such as short circuits and unstable electrical characteristics. Analyzing WAT data allows for trend analysis and prediction, enabling the development of preventive maintenance plans to extend equipment life, reduce unplanned downtime, improve productivity, and enhance wafer yields [17].

2.2.3. Process parameter data

Process Parameter Data refers to data collected in real-time by various sensors and monitoring equipment during semiconductor manufacturing [15]. The data includes numerous process parameters during the wafer testing process, such as current, voltage, chemical concentration, machine operation status, production speed, and time. Collecting and analyzing these data allows the IC manufacturing process to be monitored and controlled in real-time [18]. This data helps engineers detect and correct abnormalities promptly, reduce wafer defect density, prevent the production of unqualified products, and improve wafer yields [19].

2.2.4. Metrology data

Metrology data refers to data collected during the wafer manufacturing process. Numerous measurements are taken, including film thickness, photoresist shape, etching depth, and defect distribution [20]. Monitoring these data ensures manufacturing consistency and stability. By collecting and analyzing these data, wafer fabs can improve wafer quality, reduce the probability of failures, and thus increase yield [21].

2.2.5. CP test data

CP test data refers to the data collected after performing functional tests on each die following the completion of wafer fabrication [22]. This data typically includes various chip functional parameters. After testing, fully functional chips are sent to the next stage, which involves packaging and FT testing [23]. CP testing provides performance data for each die and the yield after CP testing.

2.2.6. FT data

FT data encompasses all test data collected during functional testing of the final manufactured IC chip [24]. It verifies the functionality and performance of each chip within design specifications. Functional tests are typically conducted during chip probing or the final test after packaging. FT data includes electrical performance testing (e.g., verifying logic gates, registers, amplification gain, filtering response) and circuit fault detection (e.g., open-circuit, short-circuit, leakage current tests) [25]. Analyzing FT data ensures chips operate according to design intentions, identify manufacturing process issues, allow for parameter adjustments, optimize production processes, and improve IC chip yields [26].

2.2.7. Data summary

Currently, there is a lack of publicly available datasets in the field of wafer yield prediction due to the high value of such datasets and the cumbersome data collection process. Additionally, there is no unified standard for data collection and processing methods—different studies employ varying data normalization and dimensionality reduction techniques. Furthermore, manual data collection may introduce fluctuations in data quality, thereby affecting the stability of model predictions. To address these challenges, future research should focus on standardizing and automating the data collection process. It is also essential to explore methods for reducing error rates in datasets or eliminating erroneous data entries to enhance dataset quality and ensure the stability of predictive models.

2.3. Wafer yield prediction

Yield is a critical metric for evaluating production efficiency and product quality in integrated circuit manufacturing [10]. Yield prediction aims to identify potential issues in the manufacturing process in advance, allowing for preventive measures to enhance the final product's pass rate [17]. Traditional yield prediction methods rely on statistical models and empirical rules [12]. However, as integrated

circuit manufacturing processes become more complex, the limitations of these methods have become increasingly apparent [15]. The introduction of AI technologies has revolutionized yield prediction.

AI-based yield prediction utilizes ML and DL techniques to analyze large volumes of historical manufacturing data, automatically identifying key factors and complex nonlinear relationships that affect yield [18]. By predicting potential issues in the manufacturing process, companies can take preventive measures to improve the final product's pass rate [20]. The general wafer yield prediction process is introduced below, as illustrated in Fig. 2.

The yield prediction process in integrated circuit manufacturing begins with data collection, which includes gathering FDC data, WAT data, process data, side data, CP test data, and FT test data. Following this, data preprocessing is conducted, involving data cleaning to remove noise and outliers [27], feature extraction to derive meaningful features such as critical process parameters and defect types [16], and data augmentation to enhance data diversity through expansion and transformation methods [10], thereby improving model generalization. The final step is model training, which entails selecting appropriate ML or DL algorithms based on data characteristics and prediction requirements [28]. Historical data is then used to train the model, followed by parameter optimization through hyperparameter tuning and cross-validation to enhance the model's prediction performance [27].

By following these steps, integrated circuit manufacturing companies can achieve accurate yield prediction, promptly identify and address production issues, and significantly improve product pass rates and efficiency. Integrating AI technology makes this process more intelligent and automated, providing strong support for companies in the competitive market [15].

In recent years, yield prediction has developed rapidly. This paper categorizes the main methods into ML and DL approaches. Machine learning methods primarily refer to techniques such as SVM [29], RF [30], and XGBoost [2], while DL methods mainly involve neural network-based approaches like LSTM [31] and ANN [32–35]. As shown in Fig. 3, this paper further classifies ML and DL methods into classification, regression, and clustering models based on the models used and the problems addressed. Classification models convert the yield prediction problem into a binary classification problem, i.e., high or low yield [30]. Regression models predict the yield value, and clustering models use unsupervised learning to cluster wafers [36] automatically. Reviewing yield prediction research from these perspectives facilitates a better understanding and analysis of current research directions and hotspots. Subsequently, this paper categorizes and introduces various ML methods based on different approaches and reviews related research.

3. Traditional machine learning methods

In the increasingly competitive semiconductor industry, wafer yield prediction has become crucial for ensuring product quality and reducing production costs [37]. Machine learning, a powerful data analysis tool, can sift through vast manufacturing data to identify patterns and correlations that traditional analysis methods may overlook [38]. By leveraging various ML algorithms, manufacturers can gain insights into the complex relationships between manufacturing processes and wafer yield, enabling them to make informed decisions to optimize production lines and improve yields [39]. This chapter introduces traditional ML models and their roles in wafer yield prediction, categorized into classification, regression, and clustering, as shown in Table 1. What is more, some abbreviations in Table 1 will be explained in Table 2.

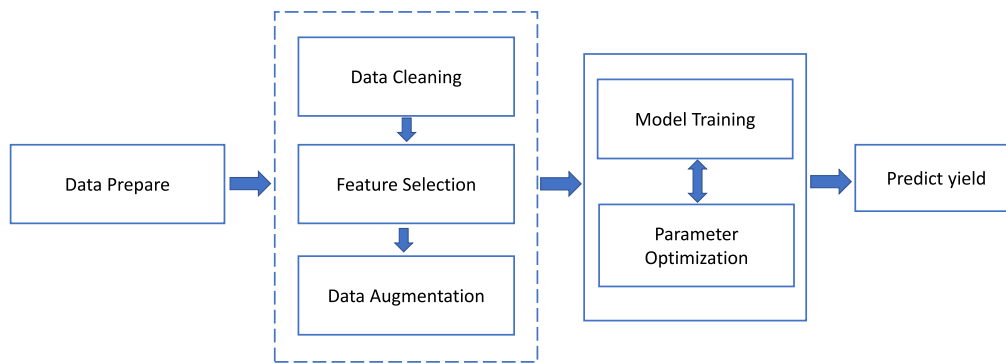


Fig. 2. Wafer yield prediction flow.

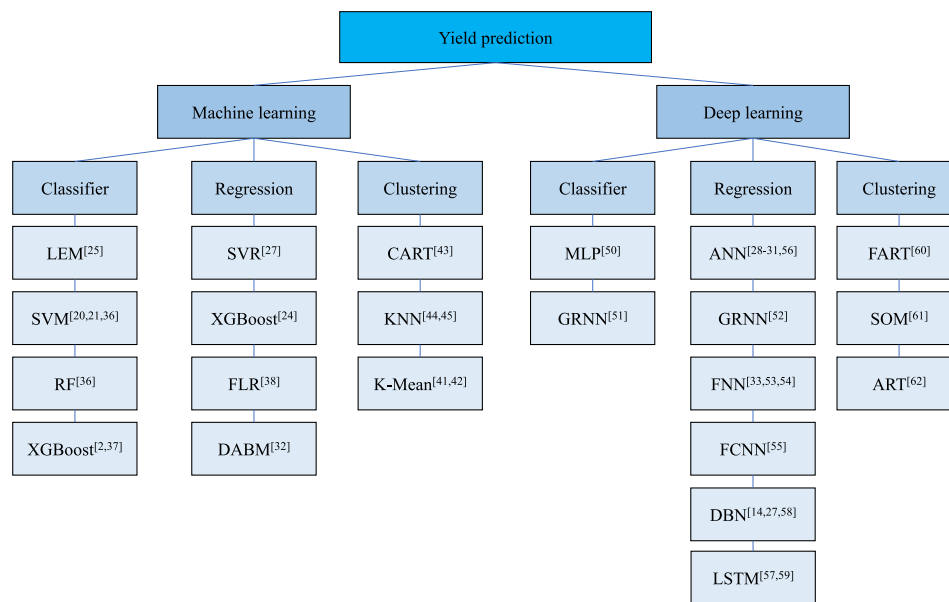


Fig. 3. Overview of wafer yield prediction methods.

3.1. Classification methods

Classification methods have become indispensable in wafer yield prediction. These methods classify wafers based on their likelihood of passing quality control tests, allowing manufacturers to identify potential defects and take corrective action quickly [38].

Fan et al. [28] proposed an integrated parameter classification method to extract process knowledge from industrial datasets for fault diagnosis and targeted optimization tasks. This scheme synthesizes mathematical ideas and decision tree techniques to derive a Linear Equipment Model (LEM). The model provides information about equipment performance and visually displays this information through graphically assisted interpretation boards, aiding in re-evaluating and optimizing control strategies. Baly et al. [24] proposed an SVM-based wafer defect classification system to improve wafer yield further. This system processes many features, classifies them efficiently, and adaptively updates the model for different engineering tasks. Experiments demonstrate that this method outperforms other baseline methods in multiple tasks, showcasing superior performance in real-world applications. For instance, SVM [40] is effective in high-dimensional spaces and is suitable for handling high-dimensional data. And it performs well when the classification boundary is clear and can handle non-linear classification problems through the use of kernel functions. Moreover, it is worth noting that SVM is sensitive to outliers and noise, necessitating data preprocessing. Park et al. [30] designed a

unique framework for yield prediction, using RF for feature selection and SVM for final result prediction. Random Forest (RF) achieves high accuracy by ensemble multiple decision trees, reducing overfitting and increasing model robustness. It is less sensitive to noise and outliers and can automatically assess the importance of each feature, facilitating feature selection, which adds to its ease of use [41]. Xu and Wang [29] employed a combination of an improved genetic algorithm, high-dimensional alternating feature selection, and SVM to handle unevenly distributed high-dimensional data for wafer yield prediction. The improved genetic algorithm addresses data sample imbalance, the high-dimensional feature selection method identifies critical parameters, and the SVM method completes the prediction. Experimental results show that this method is effective in dealing with complex datasets. However, compared to a single decision tree, RF requires longer training and prediction times, and higher memory consumption. RF provides better interpretability through feature importance evaluation, while SVM and XGBoost have poorer interpretability. SVM requires complex parameter tuning, followed by XGBoost, while RF has fewer parameters and is easier to tune [42]. Wang et al. [43] employed the PSO algorithm for feature selection and argued that while XGBoost exhibits relatively poor interpretability, it can be effectively explained using SHAP (SHapley Additive exPlanations) methods. However, class imbalance in machine learning-based yield classification — where mature fabs show high-yield dominance while R&D lines exhibit the opposite — has been addressed through GAN-based augmentation [44], genetic algorithm undersampling [29], and SMOTE techniques [45].

Lin et al. [46] used the Gaussian Process Classifier (GP) in conjunction with Laplace's method to approximate Bayesian inference. GP is selected due to its resilience to noisy data and ability to handle small wafer datasets effectively. Furthermore, models constructed using GP have the capacity to identify underlying patterns in the data, including the importance of specific input variables in predicting the wafer yield. However, the performance of GPC is sensitive to the choice of kernel function and its hyperparameters. Selecting an appropriate kernel and tuning its parameters can be challenging and often requires cross-validation or other optimization techniques, which adds to the complexity of using GPC. Song et al. [47] argue that using the average WAT values for yield prediction sacrifices accuracy. They propose partitioning wafers according to WAT test sites and performing separate yield classification predictions for each segmented region. However, it should be noted that WAT test site locations may vary across different products, which could potentially compromise the model's transferability.

3.2. Regression methods

Regression methods are widely employed to achieve higher prediction accuracy and reliability. Jiang et al. [27] introduced an innovative Gaussian Mixture Model (GMM)-based clustering ensemble regressor, which processes yield labels to enhance the accuracy of subsequent XGBoost predictions specifically for forecasting FT yield in semiconductor manufacturing processes. This approach improves prediction accuracy and enhances the model's robustness to data fluctuations by integrating the results of multiple regression models. While for large datasets, the process of clustering with GMM and then applying regression can be slow and inefficient. The need to perform multiple iterations and optimize parameters for both clustering and regression can make the approach less scalable. In many studies, yield prediction is modeled as a process influenced by various random factors within the manufacturing process, leading to significant uncertainty. This uncertainty has prompted researchers to use fuzzy methods for modeling [48]. Probabilistic or stochastic methods are considered too complex and impractical for this purpose. For example, Chen et al. [48] found that many researchers have employed Fuzzy Linear Regression (FLR) equations to model the uncertain yield learning process. This method can effectively deal with uncertainty and imprecision in data. By allowing for fuzzy input and output values, it can model real-world scenarios where data is not exact but rather expressed in terms of degrees of membership.

In the gradient boosting algorithm, prediction steps are iteratively performed using the same input feature matrix. In contrast, Lee et al. [36] proposed that the Deep Additive Boosting Model (DABM) integrates sequential prediction base models with supplemental datasets to evaluate target variables and pseudo residuals.

Moreover, various research approaches have been proposed for feature selection. Ranzel et al. [49] employed genetic algorithms for feature screening, while Zhang et al. [31] proposed a high-dimensional spatial distance measurement method using an Improved Kernel Distance Measurement (IKDM) algorithm to enable the adaptive selection of Copula functions. Through feature dimensionality reduction, model robustness can be significantly enhanced.

3.3. Clustering methods

Cluster analysis, a commonly used unsupervised learning method, classifies observations with similar characteristics [50]. Unlike classification algorithms, clustering algorithms do not assume the existence of predefined groups or clusters, making them particularly effective for analyzing unlabeled data [51].

Chien et al. [52] developed a data mining and knowledge discovery framework using k-means clustering and the Kruskal-Wallis test to analyze large amounts of wafer fabrication data, diagnosing failure

Table 1

Methods Features for Traditional Machine Learning Methods.

Type	Authors	Methods	Data	Years
Classification Methods	Fan et al.[28]	LEM	Defect Data	2001
	Baly et al.[24]	SVM	FDC&WAT	2012
	Park et al.[30]	SVM&RF	PCM	2013
	Jiang et al.[46]	GP&XGBoost	WAT&FT	2019
	Wang et al.[23]	SVM	WAT	2022
	Wang et al.[43]	XGBoost	WAT	2024
	Guo et al.[44]	CatBoost	WAT	2024
	R el al.[58]	RF	WAT	2024
Regression Methods	song et al.[47]	XGBoost&SVM	WAT	2025
	Chen et al.[48]	FLR	CP	2019
	Jiang et al.[27]	XGB	WAT&FT	2021
	Lee et al.[36,59]	RF&XGboost	WAT	2023
	Ranzel et al.[49]	RF	Process data	2023
Clustering Methods	Zhang et al.[31]	SVR	WAT	2024
	Skinner et al.[55]	CART	WAT	2002
	Wang et al.[56]	KNN	WAT	2007
	Chien et al. [52]	K-Mean	CP	2007
	Zhou et al.[57]	KNN	FDC	2014
	Hsu et al.[54]	K-Mean	FT	2015

Table 2

Abbreviations Explained in Table 1.

LEM: Linear Equipment Model	SVM: Support Vector Machine
RF: Random Forest	CART: Classification And Regression Trees
GP: Gaussian Process	XGBoost: eXtreme Gradient Boosting
MPR: Multivariate Polynomial Regression	RFR: Random Forest Regression
SVR: Support Vector Regression	KNN: K-Nearest Neighbor
DABM: Deep Additive Boosting Model	

modes and manufacturing process variations. This approach effectively identifies vital factors affecting wafer yield. K-Means scales well with the number of data points and yield features. It can handle relatively large datasets and is often used in practical applications where scalability is important. But K-Means only guarantees convergence to a local minimum, not the global optima. The final clustering result may vary between different runs due to the possibility of converging to different local minima [53].

Hsu [54] proposed a cluster integration approach to identify systematic failure modes in wafer manufacturing. He utilized the particle swarm optimization (PSO) algorithm and k-means clustering to create diverse partitions and labeled classes, then applied an adaptive response theory network for integrated results. This approach accurately identifies and classifies potential problems in the wafer fabrication process. Additionally, Skinner et al. [55] recommended using classification and regression trees (CART) to analyze yields using probe data, finding that probe data can be adequately fitted to avoid low wafer yields. The k-nearest Neighbor (KNN) clustering algorithm has been applied to multi-modal segmentation. Wang and He [56] identified multi-modal features in batch wafer fabrication processes based on KNN rules for fault detection in semiconductor manufacturing. Zhou et al. [57] addressed multi-modal identification and nonlinear problem handling in the wafer etching process using random projection and KNN rules. In conclusion, while KNN is straightforward and quick to train, it faces challenges with prediction time as datasets grow. On the other hand, K-Means requires more computational resources during training but offers better scalability for prediction tasks with large datasets.

3.4. Conclusion of chapter

Clustering methods identify potential defects by analyzing patterns in the data [60]. Classification methods build models based on known

Table 3
Methods Features for Deep Learning Methods.

Type	Authors	Methods	Data	Years
Classification Methods	Saqlain et al.[64]	MLP	PCM	2019
	Wang et al.[65]	GRNN	WAT	2019
Regression Methods	Shin et al.[32]	ANN	PCM	2000
	Tong et al.[66]	GRNN	BID	2008
	Chen et al.[67]	FNN	Field Data	2008
	Wu et al.[68]	FNN	FDC	2010
	Zhang et al.[37]	FNN	WAT&FDC	2016
	Kim et al.[69]	FCNN	Wafer map	2018
	Jang et al.[70]	ANN	Wafer map	2019
	Kim et al.[71]	LSTM	MED	2018
	Ghahramani et al.[33]	ANN	PCM	2020
	Şandru et al.[34]	ANN	PCM	2022
	Lenhard et al.[35]	ANN	PCM&FDC	2022
	Xu et al.[16,31,72]	DBN	PCM&FT	2022
	Lee et al.[73]	LSTM	FDC	2022
	Xu et al.[74]	DBN	WAT	2024
	Akbar el al.[75]	LSTM	Process data	2024
Clustering Methods	Tong et al.[76]	FART	FDC	1997
	Gardner et al.[77]	SOM	PCM&FT	2000
	Hsu et al.[78]	ART	PCM	2007

Table 4
Abbreviations Explained in Table 3.

MLP: Multilayer Perceptron	FCNN: Fully Connected Neural Network
GRNN: General Regression Neural Network	DBN: Deep Belief Network
ANN: Artificial Neural Network	FART: Fuzzy Adaptive Resonance Network
FNN: Fuzzy Neural Network	SOM: Self-organizing Feature Map
ART: Adaptive Resonance	LSTM: Long-Short Term Memory
BID: Benchmark Industrial Dataset	MED: Metrology&Equipment data

data labels to predict unknown data labels, commonly used for defect detection and quality classification [61]. On the other hand, regression methods focus on predicting continuous values, such as yield, offering the possibility of precise control and improvement of manufacturing processes [62]. Each method has features and advantages, enabling it to handle complex datasets effectively. This capability provides scientific decision support for improving wafer yield in semiconductor manufacturing processes.

4. Deep learning methods

With the advancement of AI, numerous methods have been applied to wafer yield prediction [63]. As a significant component of AI, neural networks are gaining traction across various industries, including yield prediction, due to their unique advantages. The basic information and corresponding abbreviations for all these methods are listed in Tables 3 and 4, respectively.

4.1. Classification methods

Decades ago, neural networks did not get significant attention, making it difficult to find early articles on their application in yield prediction. However, with the rise of AI in various tasks, methods for yield prediction have also advanced. Several authors [64,65] have studied wafer yield prediction using classification-based supervised learning with neural networks.

Saqlain et al. [64] integrated ML and neural networks to identify defects using a voting mechanism. Multilayer Perceptrons (MLPs) form the basic structure of the neural network, and multiple Artificial Neural Networks (ANNs) are utilized to create independent classifiers. The

classification result of a single ANN is used in a voting system to determine the outcome, incorporating results from other ANNs. This system provides quality engineers with accurate information, enabling them to take steps to improve yield. The proposed method greatly solves the limitations of traditional ML methods on non-linear datasets, however, the simple MLP architecture may also suffer from overfitting problems in practice, and even if the voting judgement method is invoked for mitigation, the existence of its own defects will reduce the reliability of the model itself. In the same year, Wang et al. [65] employed a generalized regression neural network (GRNN) with commonly used variables to build a prediction model for wafer probe yield. The validity of this model was verified using a real case from a dynamic random access memory (DRAM) manufacturer in Taiwan. The GRNN-based architecture is able to fit nonlinear relationships well and does not require complex nonlinear mapping through multiple layers of hidden cells as in MLP, which reduces the possibility of overfitting phenomena while fitting the data better.

The methods above use classification to predict yield. However, this approach may need more precision, as the output metrics of classification problems are limited to a finite number of categories and do not provide a detailed picture of the manufacturing specifics. Recognizing this drawback, many researchers and experts have sought more accurate yield prediction methods [79]. Efforts are being made to achieve breakthroughs in regression methods constructed using neural networks to enhance yield prediction accuracy [71].

4.2. Regression methods

Research on regression methods has a long history, with significant advancements since the early 20th century [80]. Neural network-based regression methods have been widely proposed for engineering applications such as wafer yield prediction, analysis, and improvement [81]. During the manufacturing process, it is crucial to ensure the accuracy of these methods while also considering features like expandability and dynamic adaptation, as higher indices in these features can lead to better yields in semiconductor manufacturing [32].

At the beginning of the 21st century, Shin et al. [32] proposed a hybrid framework combining Memory-Based Reasoning (MBR) and Neural Networks (NN) to address these issues. Neural networks can provide precise information and appropriate feature weights for optimizing semiconductor manufacturing processes or improving yields [33]. The significant increase in wafer size, driven by advanced manufacturing technology, rendered traditional mathematical-statistical models ineffective [76]. Researchers turned to neural networks to construct suitable prediction models. At the same time, the advantages unique to GRNN are such that it still has a place in the use of regression methods. In 2008, Tong et al. [66] proposed a General Regression Neural Network (GRNN) model for predicting the yield of integrated circuit wafers with clustered defects, addressing the limitations of fitting Poisson distributions to large wafer sizes. In the same year, Chen et al. [67] proposed a fuzzy neural system that combines the opinions of multiple experts. Each expert's opinion is derived from polygonal fuzzy yield prediction outputs, which are then processed by a back-propagation network to produce more accurate predictions.

Simple classification problems, such as binary classification in yield prediction, often have overly stringent boundaries, requiring direct "high" or "low" conclusions [46]. However, yield prediction in engineering does not always have clear boundaries, and fuzzy theory can help explain these fuzzy boundaries [82]. This is due to the introduction of smoothing factors in GRNN to participate in the inference process. Whether the choice of smoothing factor is reasonable or not also determines the result of inference. In practical engineering, the choice of smoothing factor is not always ideal, and fuzzy neural networks, based on fuzzy theory, have become one of the mainstays in the field of yield analysis by virtue of their strong robustness and interpretability on uncertain data. Fuzzy Neural Networks (FNN) use the concept of

affiliation to provide exact values for yield specifics. The inclusion of fuzzy rules makes FNN models interpretable. Wu et al. [68] incorporated FNN into a semiconductor manufacturing system to predict CP yield, demonstrating that FNN can capture potential relationships between independent variables and yield. In practical engineering, interference during the manufacturing process is common, necessitating adjustments to production strategies. Zhang et al. [37] proposed two FNN-based networks: one for rescheduling decisions and another for predicting yield by considering yield-influencing factors and key electrical test parameters, achieving high prediction accuracy.

Neural networks outperform traditional ML methods such as SVM and DT in yield prediction efficiency [71]. However, for application environments and data under different collection, FNN does not occupy an absolute advantage, and one of the fundamental problems of FNN is that the training volume is extremely large, and for the demand of lightweight applications, its own existence is extremely limited. At the same time, along with the dimensional characteristics of the input dataset changes, FNN's own fuzzy rules will also put forward higher requirements for DL platforms. Therefore, in practical engineering, various neural networks based on MLP architectures (e.g., fully connected networks, etc.) still occupy a place. The long and short-term time mechanisms are also introduced to capture the temporal information of various semiconductor data. Kim et al. [71] used measurement and equipment data for yield prediction, employing a model combining LSTM and a feed-forward neural network (FFNN). This approach achieved superior results compared to models such as SVR and DT. Jang et al. [70] utilized wafer map data for yield prediction, employing a carefully designed neural network as the predictive model. Kim et al. [69] constructed a deep, fully connected neural network (FCNN) to make predictions based on wafer map data. They used the geometric shapes of dies as inputs for yield prediction, achieving excellent results. Despite improvements in fitting efficiency and computational accuracy, neural networks can face overfitting or underfitting issues when dealing with large, high-dimensional data sets [83]. To address this, Ghahramani et al. [33] proposed an intelligent feature selection algorithm using Genetic Algorithms (GA) and neural networks, leveraging the cost function as a constraint to provide suitable weight distribution for the genetic algorithm, enhancing prediction effectiveness. Lee et al. [73] proposed using LSTM-AM, which combines LSTM and attention mechanisms, for yield prediction. They utilized FDC data for the predictions and employed the attention mechanism to identify factors affecting yield.

As integrated circuit (IC) production scales up, process control monitoring (PCM) data volumes increase, necessitating efficient methods [34]. From 2022 to 2023, Şandru et al. [34] and Lenhard et al. [35] developed detection models based on neural networks to improve efficiency and minimize cost. Another approach to handling large datasets is batching, which can lead to multi-modal features that help models predict relevant information [84]. It is worth mentioning that the emergence of deep belief networks provides a new way of thinking for yield analysis. This network with its adaptive feature learning, good generalization ability on multi-task makes it reliable on PCM and FT data as well. Xu et al. [72] proposed a data-driven adaptive virtual machine model based on Gath-Geva fuzzy clustering (GGFC) and multi-task learning Deep Belief Network (DBN). The DBN captures multi-modal feature information, achieving better accuracy and robustness than conventional models [85]. In addition, Xu et al. [16] first proposed a novel hybrid feature selection method that combines the mRMR method and genetic algorithm and validated the model performance using DBN. Xu et al. [31] employed copula functions for feature selection and conducted model training using SVM, BPNN, and DBN the following year. The results demonstrated that the methods based on BPNN and DBN outperformed SVM. Xu et al. [74] enhanced wafer yield prediction by implementing batch-wise analysis with a Deep Belief Network (DBN) model, coupled with a deconvolution strategy to identify critical yield-impacting parameters. This approach significantly

reduced yield improvement cycles. Akbar et al. [75] developed an LSTM-based model using process data to predict post-bonding wafer yields for the subsequent day.

In summary, AI-based methods have achieved significant yield prediction and analysis results. Neural network-based regression methods excel at capturing hidden paradigms within data and, after dimensionality reduction, can effectively capture latent information [86]. These methods address the limitations of classification algorithms by providing exact results, making neural network-based regression one of the most widely used analysis methods [87]. Typically, these methods are combined with traditional machine-learning techniques to form effective models [33]. Traditional methods provide neural networks with low-dimensional signs or relevant weight distributions, while neural networks capture hidden paradigms within the data, resulting in more accurate regression predictions [77].

4.3. Clustering methods

In the early days of semiconductor manufacturing, the lack of relevant test equipment made it impossible to obtain timely test results [88]. Consequently, improving semiconductor manufacturing yield relied heavily on unsupervised learning methods, particularly clustering techniques [76].

As discussed in Section 5.2, the increase in wafer size exacerbates defect clustering, making the Poisson distribution fit increasingly inadequate. Tong et al. [66] proposed a General Regression Neural Network (GRNN) model to address this issue, but this model was ineffective under an unsupervised learning framework. Adaptive resonance network and its variant fuzzy adaptive resonance network based on fuzzy theory, by virtue of stability, adaptive adjustment and certain memory is called one of the representatives of clustering methods. However, considering that the clustering method is only based on certain rules to group the data, and cannot fit the internal mathematical characteristics of the data more accurately, therefore, the use of regression method is often considered in yield analysis, and the use of clustering method in yield analysis has been relatively rare in recent years. In 1997, Tong et al. [76] introduced a neural network based on Fuzzy Adaptive Resonance Theory (FART), which identified the number of defect clusters and their centers, significantly reducing incorrect yield predictions caused by defective clusters.

In 2000, Gardner et al. [77] proposed a Self-organizing Feature Map (SOM) and rule induction model to identify critical undesirable factors in the wafer fabrication process. It effectively addresses issues encountered in complex manufacturing processes.

Various innovative manufacturing techniques have emerged with the advancement of semiconductor manufacturing technology, and yield improvement has become a critical reference for method selection in engineering projects [52]. The ability to locate and track process problems is vital for yield enhancement. Wafer Bin Maps (WBM) provide valuable insights for tracking process issues in semiconductor manufacturing [89]. However, most factories rely on manual visual inspection, which is time-consuming and unreliable [90]. Therefore, a reliable model or methodology is needed to track process issues through WBM to improve yields [90]. Hsu et al. [78] developed a hybrid data mining method combining spatial statistics with adaptive resonance theory neural networks. This clustering method efficiently extracts patterns from WBM and associates them with manufacturing defects [78]. Their experiments demonstrated the effectiveness and feasibility of this approach.

4.4. Conclusion of chapter

This chapter has compiled a collection of widely used neural network models for yield analysis and prediction, categorized into three sub-sections based on their underlying principles. Developing neural network-based models is a prominent research direction, demonstrating

significant effectiveness in data mining such as FDC and WAT data [59]. These studies facilitate the enhancement of subsequent models in capturing hidden paradigms within datasets, yielding more accurate results for yield prediction.

Additionally, various neural network-based methods have been applied to yield improvement, defect localization, and other related topics, achieving noteworthy results. Overall, these methods are indispensable in yield prediction, improvement, analysis, and semiconductor manufacturing [91]. It is anticipated that with the application of more efficient networks, such as Kolmogorov–Arnold Networks (KAN), Transformers, and Adversarial Training, future yield prediction and analysis methods will become increasingly efficient and precise.

5. Future research directions

Extensive research has been conducted on methods for wafer yield prediction, focusing on various aspects such as data sources and predictive models. Despite these efforts, significant gaps remain in specific research areas. A comprehensive literature review has identified notable research gaps in small sample learning, interpretable models, and multi-modal learning within the context of wafer yield prediction. This section will introduce these areas, elucidating their necessity and feasibility.

5.1. Few-shot prediction

Few-shot learning for wafer yield prediction is emerging as a crucial research direction in semiconductor manufacturing [10]. As semiconductor processes advance and product complexity increase, obtaining high-quality training data becomes increasingly complex and expensive [92]. Therefore, achieving high-accuracy yield predictions under few-shot conditions presents a key challenge. Accurate yield prediction with limited samples can save costs and improve research and development efficiency [1].

There are few studies on few-shot prediction in the yield prediction field. Busch et al. [10] addressed the issue of small data samples in yield modeling by applying multivariate non-normal distribution, correlation analysis, and principal component analysis (PCA) transformation to model variable distributions. These methods generated extensive data from small samples, enhancing the model's reliability and accuracy. Data generation methods have also been explored; for instance, Variational Autoencoders (VAEs) have been used for oversampling datasets in plasma-enhanced chemical vapor deposition processes [92]. Lee et al. [93] utilized Generative Adversarial Networks (GANs) to generate missing values in FDC data. These methods have demonstrated that data augmentation through data generation can effectively improve model performance. However, these current methods' limitations lie in their reliance on data augmentation techniques, where the quality of the generated data cannot be guaranteed. Furthermore, there needs to be more research on the models for learning from small samples, such as how to enable models to learn as much as possible from limited data.

Additionally, several techniques still need to be applied to yield prediction. By investigating these techniques, the performance of models can be enhanced, thereby overcoming the limitations above. One such technique is transfer learning, which can apply knowledge learned from one task to another related task [94]. In wafer yield prediction, models trained on large historical datasets can transfer their knowledge to new, limited sample datasets, thereby improving prediction accuracy. Semi-supervised learning combines a small amount of labeled data with many unlabeled data for training [95]. In wafer yield prediction, a small amount of labeled yield data and many unlabeled manufacturing process data can be utilized to build efficient predictive models through semi-supervised learning methods. Developing models specifically for few-shot learning, such as meta-learning and sparse learning [96,97], can quickly adapt and generalize with a small amount of training data, achieving efficient yield prediction.

Researching few-shot learning can significantly reduce data collection and labeling costs, accelerate the development cycle of new processes and products, and enhance the flexibility and competitiveness of semiconductor manufacturing.

5.2. More transparent methods

More transparent and interpretable models are emerging as an essential development direction for wafer yield prediction in semiconductor manufacturing [59]. As semiconductor processes become increasingly complex, traditional black-box models, despite sometimes providing high-accuracy predictions, often lack transparency and interpretability, which makes them less suitable for practical production needs [98]. Transparent and interpretable models not only aid in improving prediction accuracy but also provide a deeper understanding of prediction results, optimizing manufacturing processes, reducing costs, and improving yield [59].

Model interpretability techniques can enhance model transparency. For instance, methods such as Local Interpretable Model-agnostic Explanations (LIME) [99] and SHapley Additive exPlanations (SHAP) [100] can provide local explanations for each prediction result, revealing the extent to which various features influence the prediction result and making the model's decision process more transparent. Lee et al. [101] demonstrated that introducing attention mechanisms into DL models can automatically identify and focus on the most essential features for yield prediction. Their method improves the model's predictive performance and enhances its interpretability. By visualizing attention weights, engineers can intuitively identify which process parameters and steps impact yield most [101].

In the wafer manufacturing process, complex relationships exist between different process steps and parameters [98]. Graph Neural Networks (GNNs) can effectively capture these relationships and provide explanations for model predictions [102]. Engineers can better understand the interactions between process flows and parameters by visualizing the graph structure. Adopting transparent and interpretable models will enhance the reliability and efficiency of wafer yield prediction, providing valuable insights that help engineers optimize manufacturing processes and resolve potential issues [100].

5.3. Multi-modal prediction methods

In wafer fabs, various data from different sources and types, such as sensors, images, text, and time series, are available. However, current research predominantly focuses on individual data types. Some researchers, such as Wu et al. [68], have proposed using both WAT data and wafer defect data simultaneously to predict yield. They train models separately on each data type and then fuse the results to achieve better predictive performance. The primary challenge lies in effectively integrating multiple data types to enhance prediction accuracy, which involves the development of multi-modal models.

Multi-modal models integrate data from various sources and types, such as tabular, image and text, to provide more comprehensive and accurate predictions [103]. The core of multi-modal models is the effective fusion of different data types. Convolutional Neural Networks (CNNs) excel in processing image data, while Recurrent Neural Networks (RNNs) are proficient with time-series data [104]. For instance, CNNs can analyze defects in microscope images [105], and RNNs can handle time-series data from sensors [106]. Combining data from different modalities to create more complex and robust predictive models presents a challenge in multi-modal prediction. Future developments will focus on advanced data fusion techniques, including DL-based fusion methods, Bayesian networks [107], and GNNs [102]. Integrating various models and data types will become an essential research direction.

Leveraging multi-modal models will enable more comprehensive and accurate wafer yield predictions, reflecting various factors in the

manufacturing process [68]. This approach improves prediction accuracy and helps engineers understand and solve complex manufacturing issues, optimize process flows, reduce costs, and enhance product quality. Additionally, multi-modal models can be used for real-time monitoring and prediction, allowing for the prompt identification and resolution of potential issues, thereby further enhancing the efficiency and reliability of the manufacturing process [103]. In summary, multi-modal models have broad application prospects in semiconductor manufacturing, promising significant technological advancements and economic benefits to the industry.

However, in the field of integrated circuits, the acquisition, processing, and feature fusion of multimodal data pose significant challenges. First, due to the inherently high-cost nature of the industry, obtaining large-scale datasets requires substantial financial and human resources. Data privacy is another critical issue—how to build accurate predictive models while protecting industry confidentiality. Federated learning may offer a potential solution to address data privacy concerns. Second, the semiconductor industry deals with a vast variety of data types, such as the previously mentioned FDC data, WAT data, metrology data, environmental monitoring data, and more. Each wafer undergoes hundreds of manufacturing steps and dozens of testing procedures, making it extremely difficult to integrate all data into a single model. Selecting representative data is thus a key challenge. Finally, the unique characteristics of semiconductor data make feature fusion particularly challenging. Unlike other industries where image data (e.g., cats, dogs, airplanes) exhibit distinct features, semiconductor image data often shows high similarity within the same category. Even human experts may struggle to distinguish such images, let alone machines performing pixel-level analysis. Additionally, semiconductor data contains an exceptionally rich amount of information. Designing a multi-feature fusion model capable of perfectly learning all embedded information remains a major bottleneck limiting model performance.

6. Conclusions

This article comprehensively reviews recent studies on yield prediction using AI. Summarizing various ML and DL strategies offers valuable insights into the evolution of yield prediction methodologies. Additionally, analyzing the progression of yield prediction suggests potential avenues for future research. Existing yield prediction methods often face challenges such as the need for large sample sizes, lack of model transparency, and reliance on single-source data. Yield prediction also frequently encounters issues related to developing new products and increasing data diversity. Addressing these challenges represents a significant and promising direction for current and future research.

CRediT authorship contribution statement

Dongyan Zhao: Supervision, Writing – reviewing. **Yining Chen:** Methodology, Supervision, Writing – review & editing. **Pang Guo:** Conceptualization, Visualization, Investigation, Formal analysis, Writing – original draft preparation. **Fangke Chen:** Investigation, Writing – original draft preparation, Formal analysis. **Ruyue Jing:** Investigation, Writing – original draft preparation. **Xin Xu:** Writing – original draft preparation. **Yanning Chen:** Writing – review & editing. **Fang Liu:** Writing – review & editing. **Bo Wu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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